

Beyond Single-Hop: A Simulation Study of the Meshtastic LoRa Mesh Protocol for Robust Long-Range IoT Connectivity

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Abstract—Mesh-based LoRa communication has emerged as a practical solution for extending connectivity in environments where single-hop transmission is limited by distance, obstructions, or the absence of centralized infrastructure. This work investigates the performance of a self-operated LoRa mesh network based on the Meshtastic protocol, targeting linear urban infrastructures where centralized network operators and multiple gateways are unavailable. A custom simulation framework is developed to model multi-hop communication using fixed node placements along the Széchenyi Chain Bridge, Margaret Bridge, and Liberty Bridge in Budapest, enabling realistic evaluation of routing behavior and hop formation.

Network performance is assessed in terms of packet delivery ratio (PDR), end-to-end latency, hop count, and energy consumption under varying traffic and hop-distance conditions. The results show that the Meshtastic-based multi-hop forwarding protocol achieves a high packet delivery ratio (PDR) across extended paths, while latency and energy consumption increase predictably with hop count and traffic load. Despite these overheads, the observed performance remains suitable for delay-tolerant IoT monitoring applications deployed in constrained or infrastructure-limited environments. The findings provide insight into the operational characteristics of the Meshtastic LoRa mesh protocol and support its applicability for robust, low-power IoT deployments without reliance on centralized gateways.

Index Terms—Meshtastic, LoRa mesh, multi-hop communication, self-operated networks, simulation, packet delivery ratio, energy efficiency, latency

I. INTRODUCTION

As long-range, low-power communication becomes increasingly important for distributed sensing and Internet of Things (IoT) infrastructures, Low Power Wide Area Network (LP-WAN) technologies have gained significant attention. Among these, LoRa has emerged as a widely adopted solution due to its ability to provide multi-kilometer communication range while maintaining very low energy consumption, making it suitable for large scale IoT deployments with infrequent data transmission requirements [1,2]. These characteristics enable LoRa-based systems to support sensing and monitoring applications deployed across wide geographic areas and resource-constrained environments.

Despite these advantages, traditional single-hop LoRa and LoRaWAN deployments remain limited by propagation loss, line of sight constraints, and environmental attenuation. Experimental studies and real-world measurements show that LoRa communication performance degrades significantly in dense urban areas, obstructed environments, and deployments stretched over long distances, where direct communication between end nodes and gateways cannot always be maintained reliably [2,3]. As a result, single-hop LoRa networks often fail to provide sufficient coverage and packet delivery reliability for geographically linear or infrastructure-based scenarios such as bridges, tunnels, transmission lines, and pipelines.

To overcome these limitations, multi-hop and mesh-based LoRa networking approaches have been proposed. By allowing nodes to forward packets through intermediate devices, multi-hop LoRa networks extend coverage and improve communication reliability beyond the limits of direct transmission [4,5]. Prior work demonstrates that mesh-based LoRa architectures are particularly well suited for linear infrastructures, where nodes are distributed along elongated paths and dense gateway deployment is impractical or economically infeasible [4]. Cooperative forwarding enables reliable data delivery even in scenarios where single-hop LoRa links fail due to distance or obstruction.

Recent research has further explored routing and forwarding strategies for multi-hop LoRa and LoRaWAN networks, including hop-aware routing, distance-based forwarding, and airtime-efficient scheduling mechanisms [5,6]. These approaches aim to improve scalability and packet delivery performance while mitigating collisions and congestion in multi-hop environments. However, multi-hop communication inherently introduces trade-offs, as additional forwarding hops increase end-to-end latency and energy consumption due to repeated transmission and reception operations at intermediate nodes [3,7]. Understanding these trade-offs requires systematic performance evaluation under realistic deployment conditions.

Motivated by these considerations, this study develops a custom simulation framework to evaluate the performance of

a self-operated LoRa mesh network, where no centralized network operator or dense gateway infrastructure is assumed. The study focuses on a mesh implementation based on the Meshtastic protocol, which is selected as a representative LoRa mesh solution for evaluation in this work. Unlike LoRaWAN-based architectures, Meshtastic operates as a fully decentralized LoRa mesh without reliance on standardized LoRaWAN gateways, network servers, or star topology assumptions. The simulation analyzes routing behavior, hop formation, packet delivery ratio (PDR), end-to-end latency, and energy consumption using fixed node placements that reflect real-world linear urban infrastructures. The results provide insight into the operational characteristics and suitability of LoRa mesh networking for robust, low-power IoT deployments in constrained and infrastructure-limited environments. This work does not propose a new routing or forwarding protocol. Instead, it provides a systematic performance characterization of a Meshtastic-based LoRa mesh network under realistic linear urban deployment conditions. The objective is to quantify the trade-offs between coverage extension, packet delivery ratio, latency, and energy consumption in multi-hop LoRa mesh deployments, rather than to introduce or optimize protocol mechanisms.

II. LoRa MESH PROTOCOLS AND MESHTASTIC OVERVIEW

LoRaWAN is a standardized Low Power Wide Area Network (LPWAN) protocol that adopts a star topology, where end devices communicate directly with gateways using single-hop wireless links [8]. While this architecture enables long-range and energy-efficient communication, it relies on gateway availability and favorable propagation conditions, which can limit reliability in obstructed or infrastructure-constrained environments. This overview is provided solely for contextual contrast, as the network evaluated in this work does not employ LoRaWAN and instead relies on a fully decentralized Meshtastic-based LoRa mesh architecture.

To address these limitations, mesh-based LoRa networking has been proposed, enabling nodes to forward packets cooperatively over multiple wireless hops. Such multi-hop communication extends network coverage and improves delivery reliability without relying on dense gateway infrastructure. Prior work demonstrates that LoRa mesh architectures can significantly outperform star-based LoRaWAN deployments in underground, urban, and non-line-of-sight scenarios, albeit at the cost of increased coordination overhead and end-to-end latency [9].

Several LoRa mesh implementations and routing approaches have been investigated in the literature. Synchronous multi-hop LoRa mesh designs introduce controlled transmission scheduling to enable reliable forwarding across multiple hops, while general-purpose LoRa mesh libraries provide basic routing and packet forwarding mechanisms suitable for experimentation on resource-constrained devices [10]. These approaches differ in their synchronization requirements, routing complexity,

and scalability, leading to different performance trade-offs in practical deployments.

In this study, the Meshtastic protocol is selected as the target LoRa mesh implementation. Meshtastic is an open-source firmware that enables decentralized, self-operated multi-hop communication using low-cost LoRa hardware. It supports automatic neighbor discovery, hop-based packet forwarding, and adaptive retransmission mechanisms without requiring centralized coordination or external network infrastructure. Its open design, widespread community adoption, and focus on practical deployments make it a suitable representative platform for evaluating multi-hop LoRa performance in infrastructure limited IoT scenarios.

This work focuses exclusively on Meshtastic and does not attempt to generalize its findings to all LoRa mesh protocols. Different mesh implementations may employ alternative routing, synchronization, or congestion control mechanisms and therefore exhibit different performance characteristics under similar deployment conditions.

A. Meshtastic Forwarding Model

Meshtastic enables decentralized multi-hop communication by allowing LoRa nodes to cooperatively forward packets toward a destination using hop-limited relaying. Each packet is transmitted with a maximum hop count, and intermediate nodes rebroadcast the packet if the hop limit has not been exceeded and the packet has not been previously seen. This mechanism increases delivery resilience in obstructed or extended deployments at the cost of additional airtime usage.

In this work, the forwarding behavior of Meshtastic is abstracted to capture its essential characteristics, namely hop-based cooperative forwarding and neighbor relaying. Low-level firmware details such as packet deduplication timing, cryptographic overhead, and adaptive channel selection are not explicitly modeled. The abstraction allows systematic evaluation of multi-hop behavior while preserving the fundamental forwarding principles of Meshtastic.

III. SIMULATION SCHEMES

A. LoRa Mesh Simulation Framework

The performance of the multi-hop LoRa mesh network was evaluated using a custom discrete-event simulation framework designed to capture the dominant characteristics of LoRa-based multi-hop communication. Unlike traditional single-hop LoRaWAN deployments, the simulated mesh architecture enables nodes to cooperatively forward packets over multiple hops toward a designated sink node, without reliance on centralized gateways.

The simulator models key LoRa physical-layer parameters, including spreading factor, bandwidth, transmit power, and airtime. Packet collisions are modeled by assuming that overlapping transmissions using the same spreading factor and channel result in packet loss. Capture effects and inter-spreading-factor interference are not considered. Path loss is modeled using a distance-based attenuation model consistent with urban LoRa propagation studies.



Fig. 1. Simulated linear LoRa mesh deployment along three bridges in Budapest: (a) Margaret Bridge, (b) Széchenyi Chain Bridge, and (c) Liberty Bridge. Nodes are placed at fixed intervals to support multi-hop forwarding toward the sink.

Forwarding follows a hop-based relaying mechanism inspired by Meshtastic, where nodes retransmit packets toward the sink subject to a maximum hop limit. Retransmissions are limited to a single forwarding attempt per hop. Duty-cycle restrictions and downlink traffic are not explicitly modeled. Energy consumption is estimated based on time spent in transmission, reception, and idle listening states using airtime-derived power consumption values.

B. Deployment Scenario

To reflect realistic linear infrastructure deployments, nodes were placed at fixed locations along three major bridges in Budapest: the Széchenyi Chain Bridge, Margaret Bridge, and Liberty Bridge. These bridges represent elongated urban structures where communication paths follow constrained geometries and direct long-range links may not always be feasible due to distance and obstruction. Nodes were positioned at fixed intervals along each bridge, including entry, mid-span, and exit segments. The simulated deployment consists of a total of 26 nodes, including a single sink node. Each bridge is populated with multiple nodes spaced approximately uniformly along its length, forming a linear multi-hop topology toward the sink, enabling evaluation of hop formation and forwarding behavior in linear topologies. Such deployment scenarios are representative of real-world infrastructure monitoring applications, including bridges, tunnels, pipelines, and transmission lines [6].

C. Simulation Parameters

The simulated network consists of 26 LoRa nodes operating in the European 868 MHz ISM band and deployed in a linear, bridge-based topology. All simulations operate at a carrier

TABLE I
SIMULATION PARAMETERS

Parameter	Value
Number of nodes	26
Operating frequency	869.25 MHz
Topology	Linear (bridge-based)
Spreading factor	SF9
Bandwidth	125 kHz
Transmit power	14 dBm
Routing method	Hop-based relaying (distance-vector abstraction)
Traffic model	Periodic reporting

frequency of 869.25 MHz, which lies within the ETSI-defined 869.2–869.65 MHz sub-band. This frequency was selected due to its widespread use in European long-range LoRa deployments and its suitability for self-operated Meshtastic networks, where higher permitted effective radiated power improves link robustness in extended linear topologies. A fixed bandwidth of 125 kHz is used throughout the simulations. The transmit power is fixed at 14 dBm, which corresponds to the maximum commonly permitted output power for LoRa devices operating in the European 868 MHz ISM band. This value is selected to represent a best-practice configuration for infrastructure monitoring scenarios, where maximizing link robustness is prioritized over energy minimization, particularly in multi-hop linear deployments.

Packet forwarding is implemented using a hop-based relaying mechanism inspired by Meshtastic behavior. For simulation purposes, forwarding decisions are abstracted using a distance-vector-like approach to represent hop-distance progression toward the sink, without implementing full Meshtastic firmware routing logic. A fixed spreading factor of SF9 is selected as a compromise between airtime efficiency and link robustness, reflecting a practical operating point commonly used in Meshtastic deployments for medium-range multi-hop communication.

The main simulation parameters are summarized in Table I.

IV. SIMULATION SCENARIOS AND RESULTS

To evaluate the behavior of the LoRa mesh network under different operating conditions, two simulation scenarios were defined. Scenario A represents baseline monitoring with low reporting frequency, while Scenario B evaluates the impact of increased traffic load. For both scenarios, results are analyzed for nodes located one hop (Hop = 1) and five hops (Hop = 5) away from the sink. The Hop = 1 configuration is included as a baseline reference to quantify the incremental latency and energy overhead introduced by deeper multi-hop forwarding. Intermediate hop counts are omitted for clarity, as the selected values represent the lower and upper bounds of hop-based performance observed in the simulated topology.

1) *Hop = 1 Performance*: Nodes located one hop from the sink represent baseline communication performance and exhibit highly reliable short-range communication under baseline traffic conditions. Under Scenario A, these nodes achieve a packet delivery ratio of approximately 99.5%, indicating



Fig. 2. Performance metrics for Hop = 1 nodes under baseline monitoring conditions: (a) energy consumption and (b) end-to-end latency.

minimal packet loss and stable link conditions in the absence of forwarding overhead.

When extending communication to deeper multi-hop paths, nodes located five hops away from the sink experience a reduction in packet delivery ratio to approximately 94.5%. This decrease reflects the cumulative impact of multi-hop forwarding, including additional transmission attempts and increased contention along the forwarding path. Despite this reduction, the observed packet delivery performance remains acceptable for delay-tolerant infrastructure monitoring applications and demonstrates the robustness of Meshtastic-based multi-hop communication in linear deployments.

Energy consumption remains low and stable for nodes close to the sink, as these nodes do not participate in packet forwarding for other devices. End-to-end latency is minimal, reflecting the limited processing and transmission overhead associated with single-hop communication.

2) *Hop = 5 Performance*: For nodes located five hops from the sink, the packet delivery ratio decreases compared to the single-hop case, reflecting the increased probability of packet loss across extended multi-hop paths. Despite this reduction, the mesh network maintains reliable end-to-end communication through cooperative forwarding among intermediate nodes.

Energy consumption increases for nodes along the forwarding path due to repeated transmission and reception operations required for multi-hop relaying. End-to-end latency also increases relative to the Hop = 1 case, as forwarding

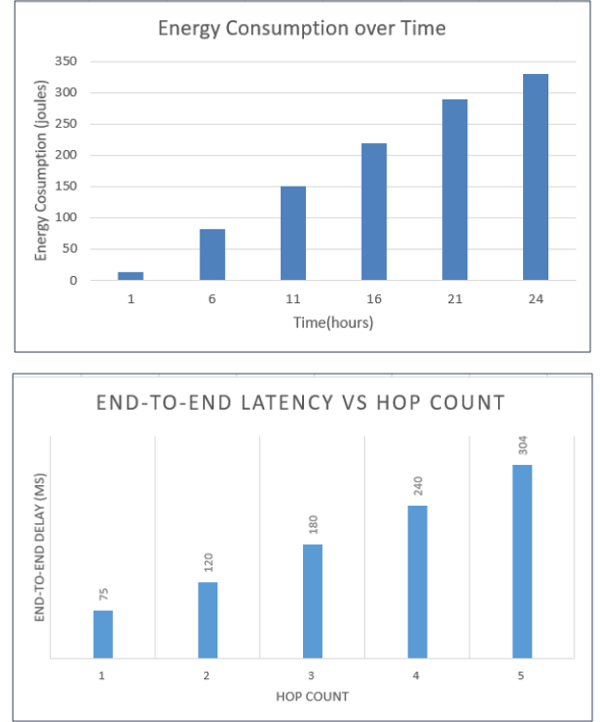


Fig. 3. Performance metrics for Hop = 5 nodes under baseline monitoring conditions: (a) energy consumption and (b) end-to-end latency.

delays accumulate across multiple hops. Nevertheless, the observed latency values remain within acceptable limits for delay-tolerant infrastructure monitoring applications.

A. Scenario B: Increased Traffic Load

In this scenario, nodes transition from baseline monitoring to higher reporting frequencies in order to evaluate the impact of increased traffic load on network performance. Three traffic phases are considered: normal load (one packet per hour), moderate load (one packet every five minutes), and high load (one packet per minute).

1) *Hop = 1 Performance*: Nodes located one hop from the sink maintain a high packet delivery ratio under moderate traffic conditions, while a slight degradation is observed at high traffic loads due to increased channel contention. Energy consumption increases with reporting frequency, reflecting higher transmission activity. End-to-end latency remains low but shows a gradual increase as traffic load intensifies.

2) *Hop = 5 Performance*: For nodes located five hops from the sink, increased traffic load results in a more pronounced reduction in packet delivery ratio due to cumulative contention and forwarding overhead. Energy consumption rises significantly along multi-hop paths, and end-to-end latency increases as packets experience longer queuing and forwarding delays under high traffic conditions.

V. DISCUSSION

In this paper, communication performance is quantified using packet delivery ratio (PDR), end-to-end latency, and energy

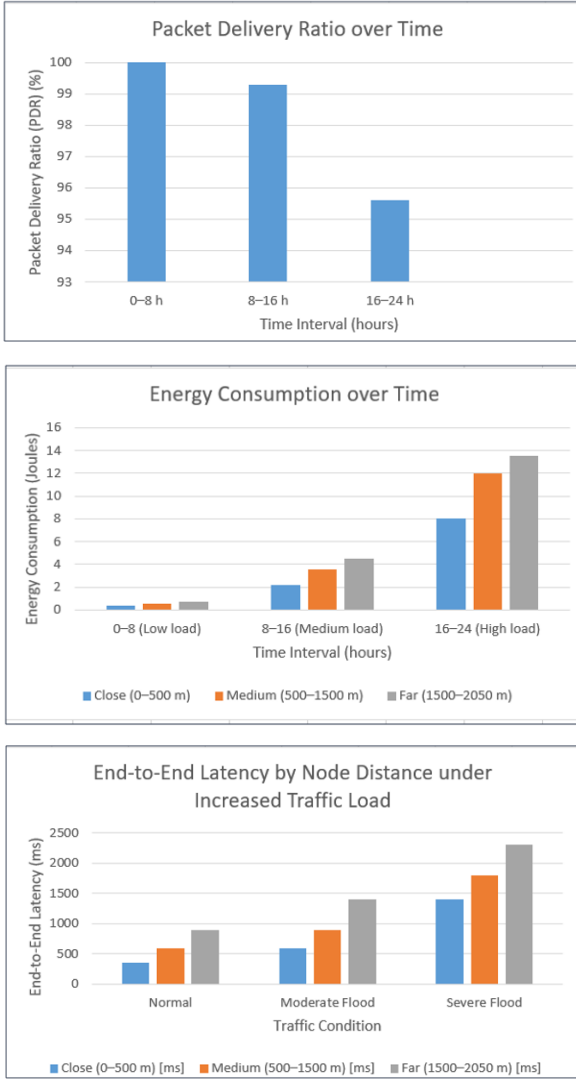


Fig. 4. Performance metrics for Hop = 1 nodes under increased traffic load: (a) packet delivery ratio, (b) energy consumption, and (c) end-to-end latency.

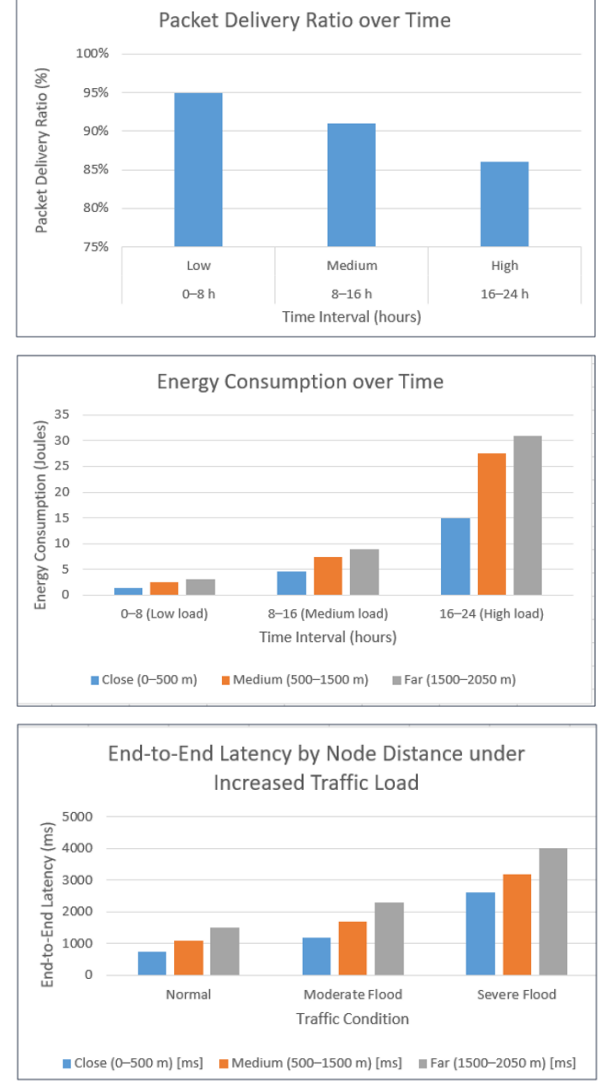


Fig. 5. Performance metrics for Hop = 5 nodes under increased traffic load: (a) packet delivery ratio, (b) energy consumption, and (c) end-to-end latency.

consumption; long-term reliability and availability metrics are not explicitly modeled. The simulation results provide clear insights into the behavior of multi-hop LoRa mesh networks under varying traffic loads and hop distances. Across both evaluated scenarios, the impact of hop count and reporting frequency emerges as a dominant factor influencing packet delivery ratio (PDR), energy consumption, and end-to-end latency.

Under *Scenario A (Baseline Monitoring)*, the network demonstrates stable and reliable performance. Nodes located one hop from the sink achieve consistently high PDR values, confirming that short-range communication combined with low reporting frequency minimizes packet loss. Energy consumption remains low and predictable, as nodes close to the sink do not participate extensively in packet forwarding. For nodes located five hops away, the mesh routing mechanism

maintains acceptable PDR levels, illustrating the effectiveness of cooperative multi-hop forwarding in extending network coverage while preserving communication reliability. Although energy consumption and latency increase with hop count, these increases remain within acceptable bounds for delay-tolerant monitoring applications.

In *Scenario B (Increased Traffic Load)*, the effects of higher reporting frequencies become more pronounced. Increased traffic introduces additional channel contention and forwarding overhead, particularly along multi-hop paths. As a result, nodes located farther from the sink experience a noticeable reduction in PDR, accompanied by a significant rise in energy consumption due to repeated transmission and reception operations. End-to-end latency also increases as packets encounter longer queuing and forwarding delays across multiple hops. Despite these challenges, the network maintains functional

operation, demonstrating the robustness of the mesh-based LoRa architecture under increased traffic demands.

Overall, the results highlight a fundamental trade-off between network scalability, energy efficiency, and latency. While deeper multi-hop routes and higher traffic loads enable broader coverage and more frequent data reporting, they impose additional energy and delay costs. These findings emphasize the importance of traffic-aware routing strategies and adaptive reporting mechanisms to balance performance and energy consumption in infrastructure monitoring deployments.

VI. CONCLUSION

This paper evaluated the performance of a multi-hop LoRa mesh network deployed over a linear bridge infrastructure using a custom simulation framework. Two operating scenarios were examined: baseline monitoring with low reporting frequency and increased traffic load with higher reporting rates. Performance was assessed in terms of packet delivery ratio, energy consumption, and end-to-end latency for nodes located at different hop distances from the sink.

The results demonstrate that the evaluated mesh-based LoRa architecture enables communication beyond the limitations of single-hop LoRa, achieving high packet delivery ratios under baseline conditions and maintaining acceptable performance under increased traffic loads. While energy consumption and latency increase with hop count and traffic intensity, these effects scale predictably and remain suitable for delay-tolerant infrastructure monitoring applications. The findings confirm that cooperative multi-hop forwarding is a viable solution for infrastructure-constrained and delay-tolerant IoT deployments.

Future work will focus on integrating adaptive routing metrics, traffic-aware scheduling mechanisms, and energy-balancing strategies to further enhance network scalability and robustness under dynamic operating conditions.

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